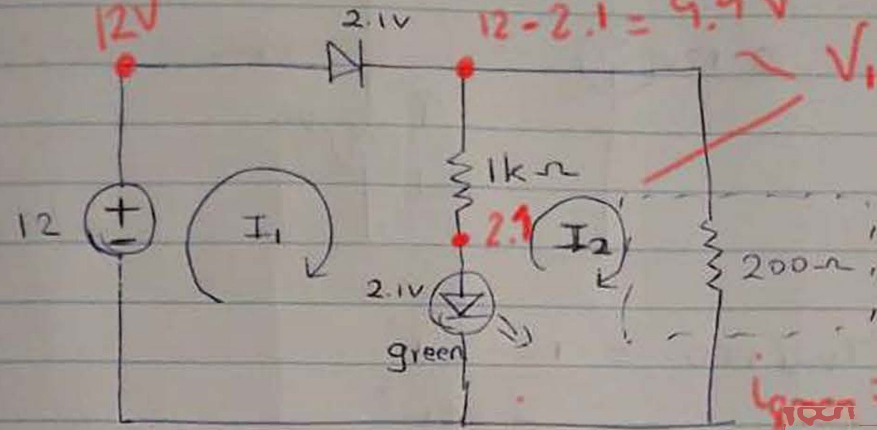
Qu1 V1

Note (from Inma): Solution in red is much easier, but many students will use mesh, so it is good to have mesh solution too.

(a)

1/ normal



$V_{1k} = 9.9 - 2.1$
$= 7.8V$

$I_{green} = \frac{7.8V}{1k} =$
$= 7.8mA$

mesh1

$$-12 + 2.1 + 2.1 + (I_1 - I_2)1000 = 0 \quad — ①$$

mesh2

$$1200 I_2 - 1000 I_1 - 2.1 = 0 \quad — ②$$

$$1000 I_1 - 1000 I_2 = 7.8 \rightarrow ①$$
$$-1000 I_1 + 1200 I_2 = 2.1 \rightarrow ②$$

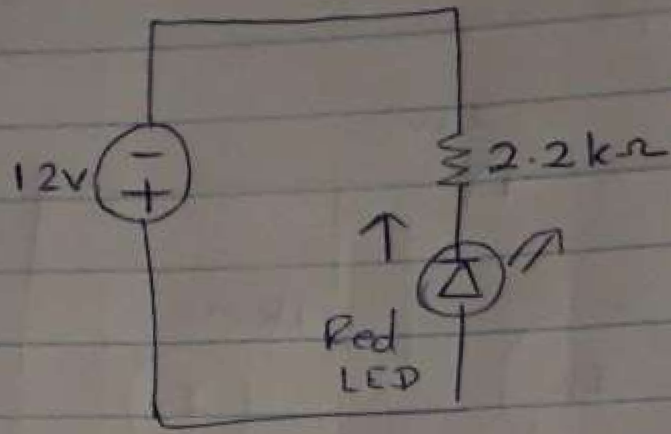
$$200 I_2 = 9.9$$
$$I_2 = 0.0495A$$

$$I_1 = 0.0573A$$

$$I_{Tx} = I_2 = 0.0495$$

$$I_{green\ LED} = I_1 - I_2 = 0.0078A$$

ii) Fault

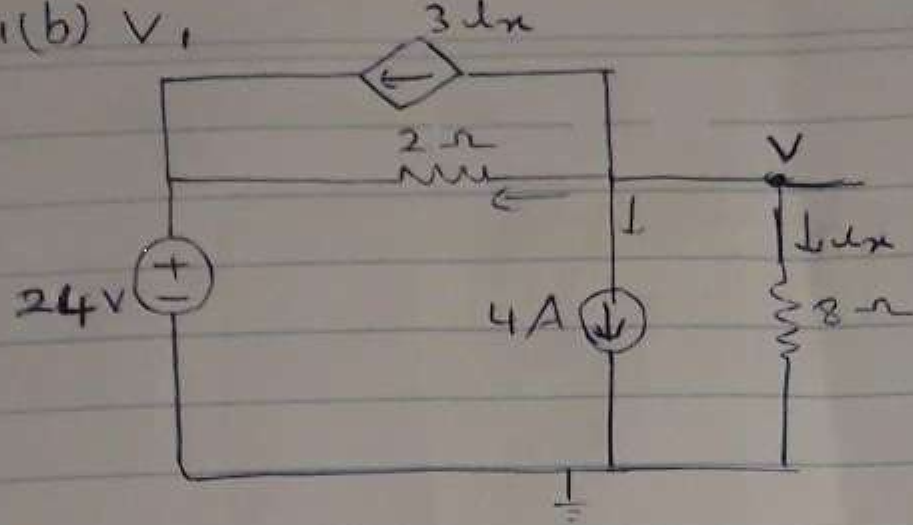


$$I_{Red\ LED} = \frac{12-2.1}{2.2\times10^{3}}$$

$$= 4.5\times10^{-3}\,A$$

$$= 4.5\,mA$$

Qu1(b) V_1



$$i_x = \frac{V}{8} \quad —①$$

nodal

$$4 + i_x + \frac{V-24}{2} + 3i_x = 0 \quad —②$$

$$② \xrightarrow{①} \quad 4 + \frac{V}{8} + \frac{V}{2} - \frac{24}{2} + \frac{3V}{8} = 0$$

$$\frac{V}{8} + \frac{V}{2} + \frac{3V}{8} = 8$$

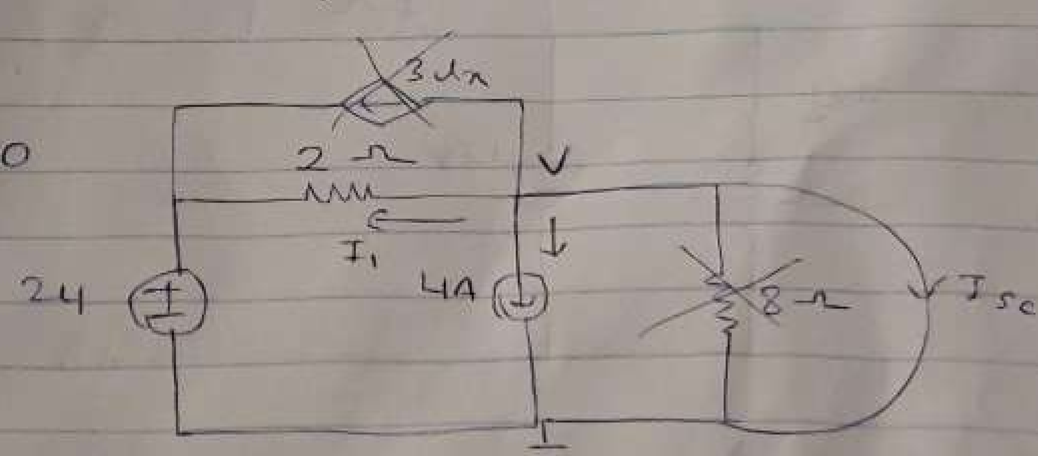
$$\frac{V + 4V + 3V}{8} = 8$$

$$8V = 8 \times 8$$

$$V = 8V$$

$V_{th} = 8V$

Isc: $i_x = 0$



$$I_{sc} + 4 + I_1 = 0$$

$$I_{sc} + 4 + \frac{V-24}{2} = 0$$

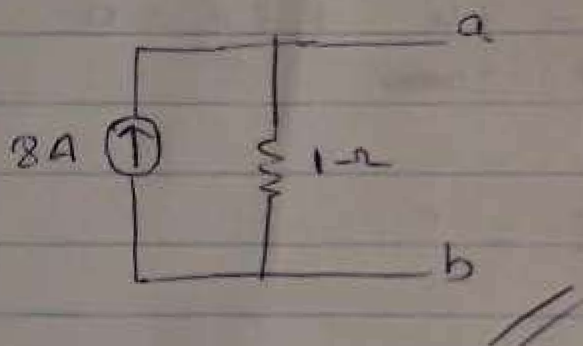
$V = 0$

$I_s + 4 - \frac{24}{2} = 0$

$I_s = 8A$ //

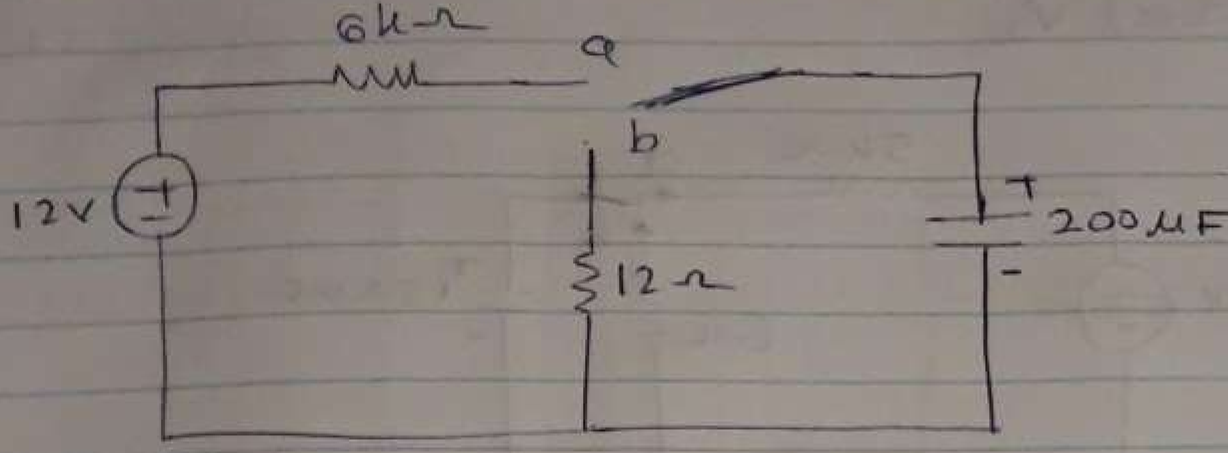
$R_{No} = \frac{V_{th}}{I_s}$

$= \frac{8}{8} = 1\Omega$



Qu2 (a)

Qu2 a) i



i/ $V_0 = 12V$

$\tau = RC = 12 \times 200 \times 10^{-6}$

$= 2400 \times 10^{-6}$

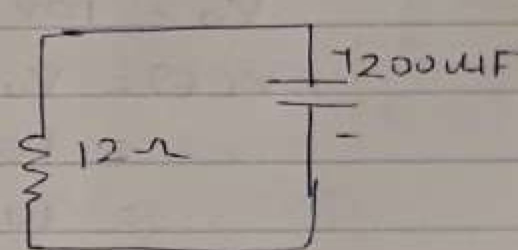
$= 0.0024$

$v(t) = V_0 e^{-t/RC}$

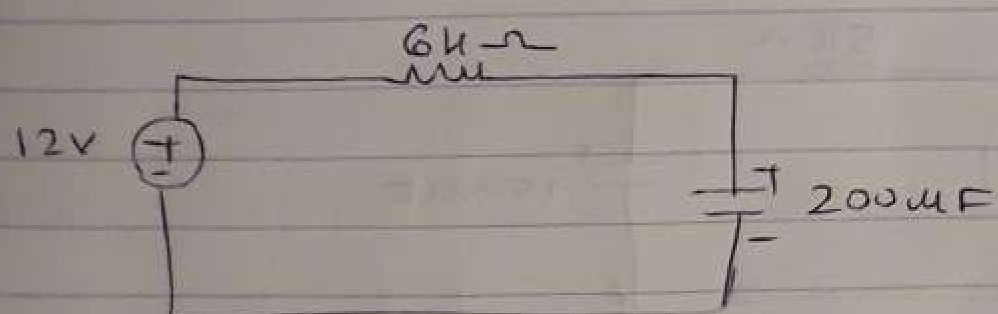
$v(t) = 12 \times e^{-t/0.0024}$

$v(t) = 12 \times e^{-\frac{0.0012}{0.0024}}$

$v(t) = 7.27V$ //



ii/



$t = 5\tau$

$= 5 \times 1.2$

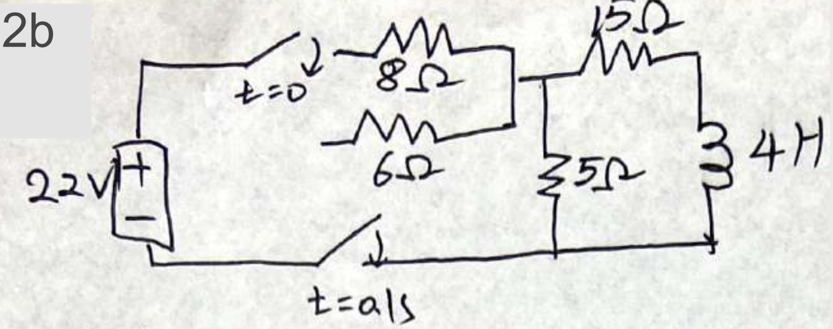
$= 6S$ //

$\tau = RC = 6 \times 10^3 \times 200 \times 10^{-6}$

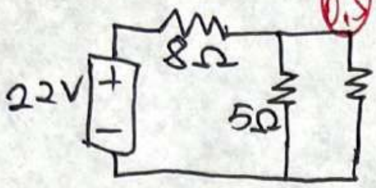
$= 12 \times 10^5 \times 10^{-6}$

$= 1.2S$

Q2b



$t<0$



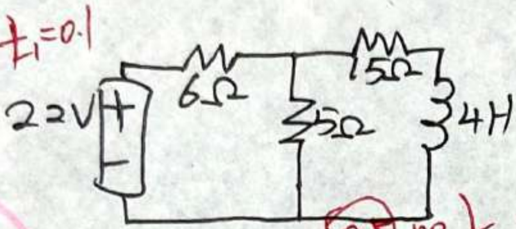
0.5 no L

$I_1 = \frac{22}{8+5//15} = 1.8723\,(A)$ 1 calculate result

$I_o(0) = I_1 * \frac{5}{20} = 0.4681\,(A)$ $\frac{22}{47}$

$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\lambda_1}$ $\frac{22}{39}$

$0<t<t_1=0.1$



0.5 no L.

steady state.

$I_1 = \frac{22}{6+5//15} = 2.2564\,(A)$ calculate result. 1

$I_o(\infty) = I_1 * \frac{5}{20} = 0.5641\,(A)$ $\frac{44}{195}$

$\lambda_1 = \frac{L}{R} = \frac{4}{15+6//5} = 0.2256$ τ 0.5

R_{th} 1

6

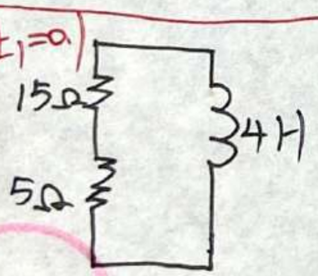
formula 0.5 $i_o(t) = i(\infty) + [i(0) - i(\infty)]e^{-t/\lambda_1}$ step

$= 0.5641 + [0.4681 - 0.5641] \cdot e^{\frac{-t}{0.2256}}$

1 Substitution

$= 0.5641 - 0.096e^{-4.4318t}$

$t>t_1=0.1$



$I_o(t) = I_o(0)e^{\frac{-(t-t_1)}{\lambda_2}}$

$i_o(0) = I_o(\infty) + [i_o(0) - I_o(\infty)] \times e^{\frac{-t_1}{\lambda_1}}$ initial condition. 1

$= 0.5641 - 0.096e^{-4.4318\times0.1}$

$= 0.5025\,(A)$

4

fraction decimal.

$\lambda_2 = \frac{L}{R} = \frac{4}{15+5} = 0.2$ 0.5 τ

$I_o(t) = I_o(0) \cdot e^{\frac{-(t-t_1)}{\lambda_2}}$ 0.5 formular.

1 $= 0.5025\,e^{(0.5-5t)}$ 0.5 time shift 0.5 Substitution

$t=t_2=0.2$

$I_o(0.2) = 0.5025\,e^{(0.5-5\times0.2)}$ 1 Substitution

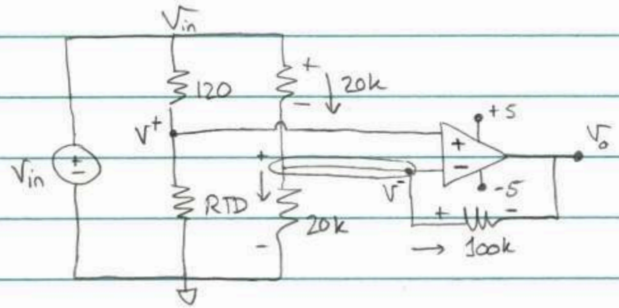
$= 0.3048\,(A)$ 1 Calculation.

2

Q3a)

(V1)

$\boxed{RTD = 130\,\Omega}$



i) $V^+ = V^- = V_{in}\,\dfrac{130}{250} = 0.52\,V_{in}$ ①

KCL @ V^-: $\dfrac{V_{in} - V^-}{20k} - \dfrac{V^-}{20k} - \dfrac{V^- - V_o}{100k} = 0$ ②

Substitute ① in ②: $\dfrac{(1-0.52)V_{in}}{2} - \dfrac{0.52V_{in}}{2} - \dfrac{0.52V_{in} - V_o}{10} = 0$

$(5\times0.48 - 5\times0.52 - 0.52)V_{in} = -V_o$

$-0.72\,V_{in} = +V_o$

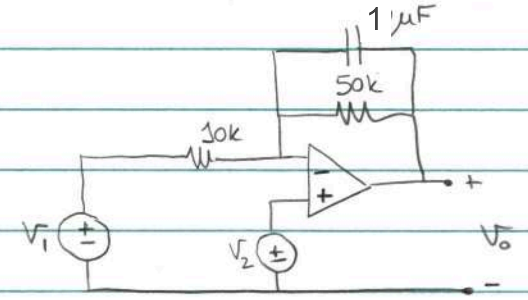
$\boxed{V_o = 0.72\,V_{in}}$

(ii) $V_o = 0.72\,V_{in} = \pm5$

$\boxed{|V_{in}| = 5/0.72 = 6.94\,V}$

Q3b)

(V1)

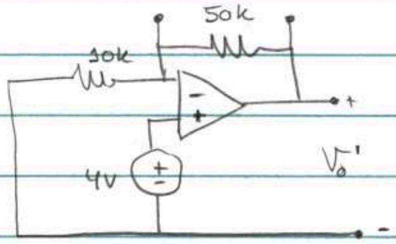


$V_1 = 3\sin(t)$ A

$V_2 = 4$V

We need to use superposition, since we have 2 sources with different frequencies

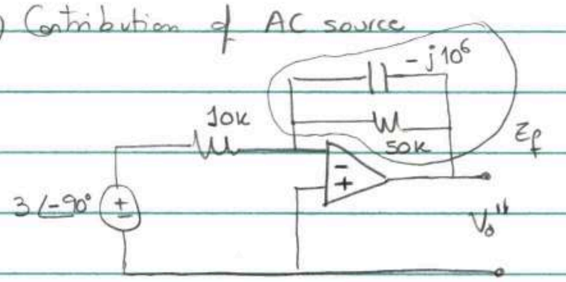
1.) Contribution of DC source



Non-inverting amplifier $V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$

$$V_o' = \left(1 + \frac{5\cancel{0}k}{1\cancel{0k}}\right) 4 = 6 \times 4 = \underline{\underline{24V}}$$

2.) Contribution of AC source



$$Z_c = -j\frac{1}{\omega C} = -j\frac{1}{1 \times 10\mu F} = -j\,10^6\ \Omega$$

Inverting amplifier $V_o = -\frac{Z_f}{Z_1} V_{in}$

$$V_o'' = \frac{-Z_f}{10k} \times 3\angle -90°$$

where $Z_f = 50k \parallel (-j10^6) = \frac{5 \times \cancel{10^4} \times (-j) \times 10^6}{5\cancel{0k} - j\,10^{\cancel{6}\,2}} =$

$$= \frac{5 \times 10^6 \angle -90°}{5 - j100} = \frac{5 \times 10^6 \angle -90°}{100.12 \angle -87.14°} =$$

$$= 49940.07 \angle -2.86°\ \Omega$$

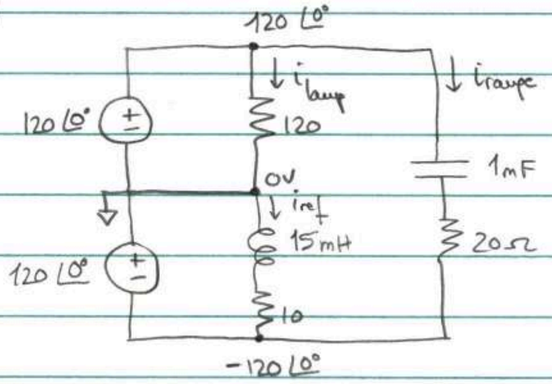
$$V_o'' = \frac{49940.07 \angle 180 - 2.86° \times 3\angle -90°}{10k} = \underline{\underline{14.98 \angle 87.14°\ V}}$$

3.) Add contributions in time domain

$$\underline{V_o(t) = 24 + 14.98\cos(t + 87.14°)\ V}$$

Q4 a)

V1



First, convert to phasor domain ($f = 60Hz$)

$$Z_L = j\omega L = j \times 2\pi 60 \times 15 \times 10^{-3} = j\,5.65\ \Omega$$

$$Z_C = -j\frac{1}{\omega C} = -j\frac{1}{2\pi 60 \times 10^{-3}} = -j\,2.65\ \Omega$$

Then, calculate currents:

$$I_{lamp} = \frac{120\angle 0^\circ}{120} = 1\text{ A}$$

$$I_{ref} = \frac{120\angle 0^\circ}{10 + j5.65} = \frac{120\angle 0^\circ}{11.486\angle 29.47^\circ} = 10.45\angle -29.47^\circ\ A$$

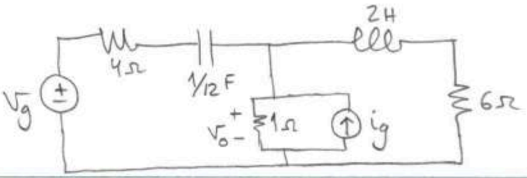
$$I_{range} = \frac{120 - (-120)}{20 - j2.65} = \frac{240\angle 0^\circ}{20.175\angle -7.55} = 11.9\angle 7.55^\circ\ A$$

Finally, convert back to time domain:

$$i_{lamp} = \cos(120\pi t)\ A$$

$$i_{ref} = 10.45\cos(120\pi t - 29.47^\circ)\ A$$

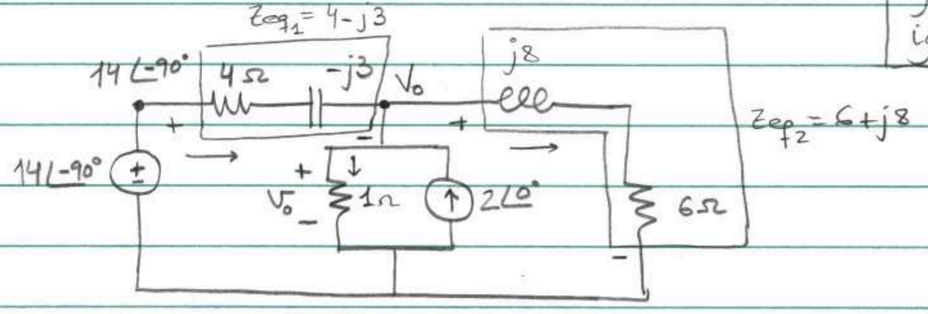
$$i_{range} = 11.9\cos(120\pi t + 7.55^\circ)\ A$$



$V_g = 14 \sin 4t$ V
$i_g = 2 \cos 4t$ A

Q4 b)

(V1)



First, convert into phasor domain ($\omega = 4$ rad/s)

$v(t) = 14 \sin 4t \longrightarrow V = 14\angle{-90°}$

$i(t) = 2 \cos 4t \longrightarrow I = 2\angle{0°}$

$Z_C = -j \dfrac{1}{\omega C} = -j \dfrac{1}{4 \times 1/12} = -j3\,\Omega$

$Z_L = j\omega L = j\,4 \times 2 = j8\,\Omega$

Then, use nodal analysis to find V_o:

$$\frac{14\angle{-90°} - V_o}{4 - j3} - V_o + 2 - \frac{V_o}{6 + j8} = 0$$

$$+\left(\frac{+1}{4-j3} + 1 + \frac{1}{6+j8}\right) V_o = +\left(2 + \frac{14\angle{-90°}}{4 - j3}\right)$$

$$\left(1 + \frac{1}{5\angle{-36.87°}} + \frac{1}{10\angle{53.13}}\right) V_o = 2 + \frac{14\angle{-90°}}{5\angle{-36.87°}}$$

$$\left(1 + 0.2\angle{36.87} + 0.1\angle{-53.13}\right) V_o = 2 + 2.8\angle{-53.13}$$

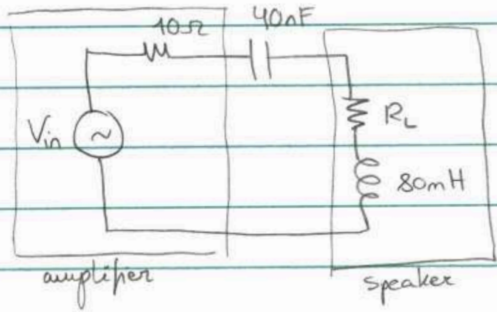
$$V_o = \frac{2 + 1.68 - j2.24}{1 + 0.16 + j0.12 + 0.06 - j0.08} = \frac{3.68 - j2.24}{1.22 + j0.04} = \frac{4.31\angle{-31.33}}{1.22\angle{1.88°}} = 3.53\angle{-33.21°} \text{ V}$$

Finally, convert back to time domain:

$v_o(t) = 3.53 \cos(4t - 33.21°)$ V

Q5 a)

v1



Power in speaker

$|V_{in}| = 4.6\ V_{rms}$

i) For MAPT $Z_L = Z_{Th}^*$, where $Z_L = R_L + j\omega L$

$$Z_{Th} = 10 - j\frac{1}{\omega C}$$

$$R_L = R_{Th} \Rightarrow \boxed{R_L = 10\Omega}$$

$$X_L = -X_{Th} \Rightarrow \cancel{j}\omega L = \cancel{j}\frac{1}{\omega C}$$

$$\omega^2 LC = 1 \Rightarrow \boxed{\omega = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{80\times10^{-3}\times40\times10^{-9}}} = 17677.67\ \text{rad/s}}$$

ii) $V_{in} = 4.6\ V_{rms}$

Note: $P_{max} = \frac{|V_{Th}|^2}{8R_{Th}}$ will give same result in questions for power in speaker

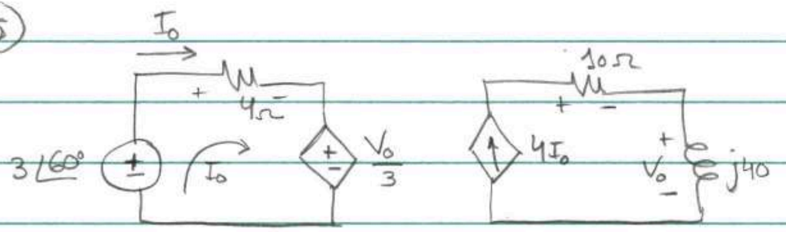
$$\boxed{P_{speaker} = P_{10\Omega} + \underbrace{P_{80mH}}_{0} = \frac{V_{10\Omega\,rms}^2}{10} = \frac{2.3^2}{10} = 0.529\ W}$$

Use voltage division to find $V_{10\Omega\,rms}$: $V_{10\Omega\,rms} = V_{in}\frac{10}{20} = 2.3\ V_{rms}$

Q5 b)

v5

$\boxed{P_{avg} \text{ in } 10\Omega \text{ resistor}}$



$$P_{10\Omega} = \frac{1}{2} V_{m_{10}} I_{m_{10}} = \frac{1}{2} \frac{V_{m_{10}}^2}{10} = \frac{1}{2} I_{m_{10}}^2 \times 10$$

$$I_{10\Omega} = 4I_o \quad ①$$

Find equation for I_o from left mesh:

(KVL) $-3\angle 60° + 4I_o + \frac{V_o}{3} = 0$ ②, where $V_o = j40 \times 4I_o = j160 I_o$

$$-3\angle 60° + 4I_o + j53.3 I_o = 0$$

$$I_o = \frac{+3\angle 60°}{4 + j53.3} = \frac{3\angle 60°}{53.45\angle 85.71°} = 0.056\angle -25.71°$$

From ①, $I_{10\Omega} = 4I_o = 0.224\angle -25.71°\,A$ $\Rightarrow$ $\boxed{P_{10\Omega} = \frac{1}{2} I_m^2 \times 10 = 5 \times 0.224^2 = 0.251\,W}$